

SQL Server Database Migration – On-prem to Azure SQL Fundamentals

1. What is Database Migration?

Topic	Easy Explanation
Database Migration	Moving a database from one system/server to another
Common Scenario	Moving on-prem SQL Server databases to Azure SQL or newer SQL Server
Goal	Improve performance, scalability, security, and maintenance
Migration Types	Homogeneous (SQL → SQL) and Heterogeneous (Oracle/MySQL/Access → SQL Server)
Main Challenge	Compatibility, downtime, schema conversion, and data integrity

2. Types of SQL Server Migrations

Migration Type	Description	Example
SQL Server → SQL Server	Upgrade or move SQL instances	SQL 2016 → SQL 2022
SQL Server → Azure SQL Database	Cloud migration	On-prem → Azure SQL
SQL Server → Azure VM	Move full SQL Server into Azure VM	Lift-and-shift
Non-SQL → SQL Server	Convert other DB engines	Oracle → SQL Server
Access → SQL Server	Upsize Access DB	MS Access → SQL Server

3. Why Organizations Migrate Databases

Reason	Explanation
Better Performance	Faster query execution
Scalability	Handle larger databases and more users
Cloud Adoption	Reduce hardware management
Security	Advanced authentication and encryption
High Availability	Disaster recovery and failover support
Lower Maintenance	Azure handles patches and backups
Modernization	Replace legacy infrastructure

4. Migration Strategies

Strategy	Meaning	Downtime
Offline Migration	Stop application during migration	High
Online Migration	Continuous replication during migration	Low
Lift and Shift	Move as-is without redesign	Medium
Re-platform	Small modifications for cloud	Medium
Refactor	Redesign application/database	High

5. Microsoft Migration Tools

Tool	Purpose	Usage
SQL Server Migration Assistant (SSMA)	Converts non-SQL databases	Oracle/MySQL/Access
Azure Database Migration Service (DMS)	Online/offline migration service	Azure migrations
Data Migration Assistant (DMA)	Compatibility assessment	Pre-migration analysis
Azure Data Studio Migration Extension	Guided migration workflow	Azure SQL
SQL Server Management Studio (SSMS)	Backup/restore and migration operations	SQL admin tasks

6. Migration Workflow (Typical Process)

Phase	Tasks
Assessment	Check compatibility and blockers
Planning	Choose target platform and downtime window
Schema Migration	Move tables, indexes, procedures
Data Migration	Transfer actual data
Validation	Compare source and target
Cutover	Switch application to new DB
Optimization	Tune indexes and performance

7. Pre-Migration Assessment

Check	Why Important
Compatibility	Some SQL features may not work in Azure
Data Types	Unsupported data types cause failures
Deprecated Features	Old SQL syntax may break
Large Tables	Need special migration planning
Security Objects	Logins/users/roles must be migrated
SQL Agent Jobs	Jobs do not always migrate automatically

8. Azure SQL Options

Azure Option	Best For	Advantages	Limitations
Azure SQL Database	Modern cloud apps	Fully managed	Limited instance-level features
Azure SQL Managed Instance	Near full SQL compatibility	Easy migration	Higher cost
SQL Server on Azure VM	Lift-and-shift	Full SQL control	Manual management

9. Offline vs Online Migration

Feature	Offline	Online
Downtime	Required	Minimal
Complexity	Simpler	More complex
Data Sync	One-time copy	Continuous replication

Feature	Offline	Online
Best For	Small DBs	Production critical systems
Speed	Faster setup	Longer preparation

10. Common Migration Methods

Method	Description	Best Use
Backup & Restore	Restore DB backup on target	SQL → SQL
Detach & Attach	Move MDF/LDF files	Same environment
BACPAC Export	Export schema + data	Azure SQL
Transactional Replication	Near real-time sync	Minimal downtime
Log Shipping	Restore transaction logs continuously	DR and migration
SSIS	ETL-based migration	Complex transformations

11. SQL Server to Azure SQL Migration

Step	Action
1	Assess with DMA
2	Fix compatibility issues
3	Create Azure SQL target
4	Configure Azure DMS
5	Start migration
6	Validate data
7	Perform cutover

12. Non-SQL to SQL Migration

Source DB	Tool Used
Oracle	SSMA for Oracle
MySQL	SSMA for MySQL
DB2	SSMA for DB2
Sybase	SSMA for Sybase
Access	SSMA for Access

13. Access to SQL Server Migration Notes

Topic	Important Point
Split Database	Front-end and back-end separation helps
Primary Keys	Every table should have PK
Indexes	Required for performance
Attachment Columns	Not migrated properly

Topic	Important Point
Linked Tables	Access front-end can still connect to SQL Server
Data Types	Must validate carefully

14. Migration Risks

Risk	Impact
Data Loss	Missing records after migration
Downtime	Business interruption
Compatibility Issues	Application failures
Performance Regression	Slow queries
Security Misconfiguration	Unauthorized access
Broken Jobs/Triggers	Automation failures

15. Best Practices

Best Practice	Reason
Run Test Migration	Detect issues early
Backup Everything	Recovery safety
Validate Row Counts	Ensure data integrity
Use Staging Environment	Avoid production failures
Monitor Performance	Tune after migration
Document Dependencies	Prevent missing components
Migrate Logins Separately	Users may break otherwise

16. Migration Architecture Overview

Component	Role
Source Database	Existing production DB
Assessment Tool	Checks compatibility
Migration Service	Transfers schema/data
Target Database	New SQL/Azure DB
Validation Layer	Confirms successful migration
Application Cutover	Redirect apps to new DB

17. SQL Server Migration Assistant (SSMA)

Feature	Description
Schema Conversion	Converts source schema to SQL Server
SQL Translation	Converts SQL syntax
Data Migration	Transfers records

Feature	Description
Assessment Reports	Shows incompatibilities
Supported Sources	Oracle, MySQL, DB2, Sybase, Access

18. Azure Database Migration Service (DMS)

Feature	Description
Managed Azure Service	Microsoft-hosted migration engine
Online Migration	Minimal downtime
Multiple Sources	SQL, Oracle, MySQL
Monitoring	Migration progress tracking
Hybrid Support	On-prem to cloud

19. Key Migration Terminology

Term	Meaning
Schema	Structure of database
ETL	Extract, Transform, Load
Cutover	Final switch to new DB
Replication	Continuous data sync
Compatibility Level	SQL feature support mode
Latency	Delay in synchronization
HA/DR	High Availability / Disaster Recovery

20. Real-World Practical Advice

Advice	Why It Matters
Dry-run migrations	Prevent production surprises
Migrate during low traffic	Reduce business impact
Test applications after migration	DB migration alone is insufficient
Pre-size tempdb and storage	Avoid performance bottlenecks
Validate stored procedures	Encrypted or legacy SPs may fail

21. Common Interview Questions

Question	Short Answer
What is database migration?	Moving DB between systems/platforms
Difference between online and offline migration?	Downtime vs near-zero downtime
What is SSMA?	Tool for converting non-SQL DBs
What is DMS?	Azure migration service
Why use Azure SQL Managed Instance?	High SQL Server compatibility
What is cutover?	Final switch to target DB

22. One-Line Summary

Topic	Summary
SQL Migration	Process of moving and modernizing databases safely
Main Microsoft Tools	SSMA, DMA, DMS, Azure Data Studio
Biggest Challenges	Compatibility, downtime, validation
Best Approach	Assess → Test → Migrate → Validate → Cutover

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